Enrollment No.:.....

Signature of the Student:.....

MAHATMA GANDHI CENTRAL UNIVERSITY, MOTIHARI (BIHAR)

Engineering Graphics

B.Tech. I sem Class Test-I, 2025

Computer Science and Information Technology

Maximum Marks: 7.5 Time Allowed: 30 Minutes

1. **A device which combines the functions of a T-square, set square, protractor and scale is called**
   (a) fasteners (b) mini drafter (c) templates (d) combination set
2. **In the engineering system of paper sizes, which of the following is 'A2' size?**
   (a) 841 mm × 1189 mm (b) 594 mm × 841 mm
   (c) 420 mm × 594 mm (d) 210 mm × 297 mm
3. **Which of the following pencil leads is hardest?**
   (a) HB (b) H (c) B (d) F
4. **To draw smooth curves of any nature, the drafting instrument used is a/an**
   (a) mini drafter (b) French curve (c) template (d) eraser shield
5. **Paper size 'A0' has an area of**
   (a) 1 m^2 (b) 2 m^2 (c) 3 m^2 (d) 4 m^2
6. **Centre lines are drawn as**
   (a) continuous narrow lines (b) dashed narrow line
   (c) long-dashed dotted narrow line (d) long-dashed double dotted narrow line
7. **Parallel lines can be drawn with the help of**
   (a) mini drafter (b) T-square (c) pair of set-squares (d) All of these
8. **Hidden lines are drawn as**
   (a) dashed narrow line (b) dashed wide line
   (c) long-dashed dotted narrow line (d) long-dashed dotted wide line
9. **The dimension value for diameter of a circle should be**
   (a) preceded by the symbol 'Ø' (b) succeeded by the symbol 'Ø'
   (c) preceded by the letter 'D' (d) succeeded by the letter 'D'
10. **For drawing the components of a wrist watch, the scale used is**
    (a) reduction scale (b) full size scale (c) enlargement scale (d) Any of these
11. **Drawing of a building can be made on A3 size drawing sheet using**
    (a) reduction scale (b) full size scale (c) enlargement scale (d) None of these
12. **The R.F. is always**
    (a) less than 1 (b) equal to 1 (c) greater than 1 (d) Any of these
13. **The length of scale with R.F. 1/40 to measure up to 6 m will be**
    (a) 10 cm (b) 12 cm (c) 15 cm (d) 20 cm

B.Tech. I sem Class Test-II, 2025

Computer Science and Information Technology

Maximum Marks: 7.5

Time Allowed: 30 Minutes

1. **A curve generated by a point lying on the circumference of a circle which rolls along a fixed straight or curved path without slipping is called**
   (a) cycloidal curve (b) trochoidal curve (c) involute (d) spiral
2. **The curve traced out by a point on the circumference of a circle which rolls on another circle with external contact is**
   (a) epicycloids (b) hypocycloid (c) superior epitrochoid (d) superior hypotrochoid
3. **The curve traced out by a point on the circumference of a circle which rolls on another circle with internal contact is**
   (a) epicycloids (b) hypocycloid (c) superior epitrochoid (d) superior hypotrochoid
4. **An object shown by more than one views in a drawing is called**
   (a) perspective drawing (b) isometric drawing (c) oblique drawing (d) multi-view drawing
5. **What is the reference line (XY) in projection drawings?**
   (a) The ground level (b) The line of intersection between HP and VP
   (c) The object's edge (d) A dimension line
6. **Which of the following describes the theory of orthographic projections?**
   (a) Projectors are parallel to each other and per pendicular to the plane of projection. (b) Projectors are parallel to each other and parallel to the plane of projection.
   (c) Projectors are parallel to each other and oblique to the plane of projection.
   (d) Projectors are perpendicular to each other and parallel to the plane of projection.
7. **In orthographic projection, how many standard views are typically used to fully describe a simple object?**
   a) 1 b) 2 c) 3 d) 4
8. **What does the "front view" in orthographic projection show?**
   (e) The top surface of the object
   (f) The view as if looking straight at the object's front face
   (g) The side profile only
   (h) A bird's-eye perspective
9. **Which angle projection places the object in the first quadrant, with the top view below the front view?**
   (a) Third-angle projection (b) First-angle projection
   c) Isometric projection d) Perspective projection
10. **What does the "top view" in orthographic projection show?**
    (a) The view as if looking straight at the object's front face
    (b) The top surface of the object (c) The side profile only (d) A bird's-eye perspective

11. **Why are hidden lines drawn in orthographic projection?**
(a) To show colors (b) To indicate edges not visible in that view
(c) To make the drawing colorful (d) To measure angles

12. **In first-angle projection, where is the top view placed relative to the front view?**
(a) Above it (b) Below it (c) To the right (d) To the left

13. **Why is orthographic projection called "multiview" drawing?**
a) It uses multiple colors b) It shows the object from multiple directions
c) It requires multiple drawings d) It has multiple layers

14. **If a point lies on the horizontal plane (HP), its top view will be:**
(a) Above the XY line (b) On the XY line (c) Below the XY line d) Not visible

15. **If a line is parallel to both HP and VP, its projections on both planes will be:**
a) Points b) True length lines c) Perpendicular to each other d) Invisible

Q No.	Correct option	
1	a	
2	a	
3	b	
4	d	
5		
6		
7	c	
8	b	
9	b	
10	b	
11		
12		
13		

Name of the Student: PIYUSH SINHA

Enrollment No. MGCU2024CSIT3034

Signature of the Student: Piyushsinha

MAHATMA GANDHI CENTRAL UNIVERSITY, MOTIHARI (BIHAR)

Engineering Graphics

B.Tech. I sem Mid Sem Exam, 2025

Computer Science and Information Technology

Maximum Marks: 20 Time Allowed: 2 Hour

Section A

1. **Choose the correct answer of the following (any five):** *[1 mark x 5= 5 Marks]*

   i. In the engineering system of paper sizes, which of the following is 'A2' size?

   (a) 841 mm × 1189 mm (b) 594 mm × 841 mm

   (c) 420 mm × 594 mm (d) 210 mm × 297 mm

   ii. A device which combines the functions of a T-square, set square, protractor and scale is called

   (a) fasteners (b) mini drafter

   (c) templates (d) combination set

   iii. A drafter helps in drawing

   (a) parallel and perpendicular lines (b) concentric circles

   (c) smooth curves (d) All the these

   iv. Long-dashed dotted narrow line is used to represent

   (a) line of symmetry (b) centre lines

   (c) pitch circle of gears and holes (d) All of these

   v. The length-to-height ratio of a closed filled arrow head is

   (a) 1:3 (b) 3:1 (c) 1:2 (d) 2:1

   vi. The curve traced out by a point on the circumference of a circle which rolls on another circle with internal contact is

   (a) epicycloids (b) hypocycloid (c) superior epitrochoid (d) superior hypotrochoid

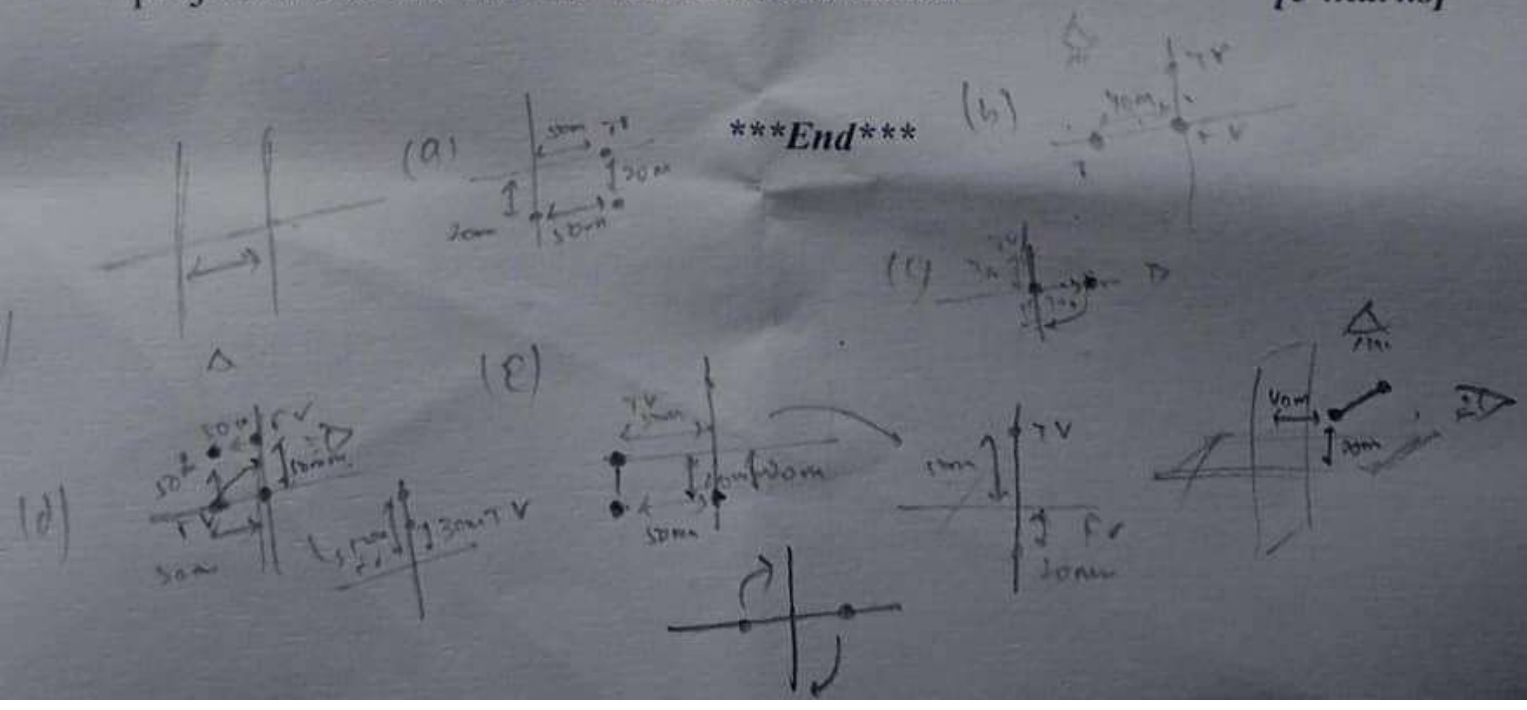
Section B

1. The distance between Mahatma Gandhi Central University, Motihari (Bihar) campus and Central University of South Bihar (Gaya) by road is 250 kilometres. It is represented on a certain map by a 5 centimetres long line. Find the representative fraction (RF). *[2.5 marks]*

2. Draw the projections of the following points on a common reference line keeping the distance between their projectors 30 mm apart. (a) Point A is 20 mm below the H.P. and 50 mm in front of the V.P. (b) Point B is in the H.P. and 40 mm behind the V.P. Projections of Points 8.9 (c) Point C is 30 mm in front of the V.P. and in the H.P. (d) Point D is 50 mm above the H.P. and 30 mm behind the V.P. (e) Point E is 20 mm below the H.P. and 50 mm behind the V.P. *[2.5 marks]*

Section C

3. Construct a scale of 1:40 to read metres and decimetres and long enough to measure up to 6 metres. Mark a distance of 4.7 m on it. *[5 marks]*

4. A 80 mm long line PQ has end P 20 mm above H.P. and 40 mm in front of the V.P. The line is inclined at 30° to the H.P. and is parallel to the V.P. Draw the projections of the line and determine its traces. *[5 marks]*

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